

Challenge 1 Rosettes and Spirals

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# RosettesAndSpirals.py - a spiral of shapes AND rosettes!

import turtle

t=turtle.Pen()

t.penup()

t.speed(0)

turtle.bgcolor('black')

# Ask the user for the number of sides, default to 4, min 2, max 6
sides = int(turtle.numinput("Number of sides",
"How many sides in your spiral?", 4,2,6))

colors=["red", "yellow", "blue", "green", "purple", "orange"]

# Our outer spiral loop
for m in range(1,60):

    t.forward(m*4)

    t.width(m//25+1)

    position = t.position() # remember this corner of the spiral

    heading = t.heading() # remember the direction we were heading

    # Our 'inner' spiral loop,

    # draws a little rosette at each corner of the big spiral

    if (m % 2 == 0):

        for n in range(sides):

            t.pendown()

            t.pencolor(colors[n%sides])

            t.circle(m/4)

            t.right(360/sides - 2)

        t.penup()
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# OR, draws a little spiral at each corner of the big spiral
else:
    for n in range(3,m):
        t.pendown()
        t.pencolor(colors[n%sides])
        t.forward(n*2/3)
        t.right(360/sides - 2)
        t.penup()
        t.setpos(position) # reset the position on the big spiral
        t.setheading(heading) # point in the big spirals direction/heading
        t.left(360/sides + 2) # move to the next point on the big spiral
```